Pion–proton identification and cluster-level energy reconstruction in a semi-digital hadron calorimeter

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Abstract

Highly granular semi-digital calorimetry provides both threshold information and a three-dimensional description of hadronic showers. We study how these observables can be used for binary identification of negative pions and protons and for single-hadron energy reconstruction in a simulated 48-layer glass resistive-plate chamber calorimeter. Two separate Geant4 productions are used: a continuous sample between 1 and 130 GeV for training and calibration, and a fixed-energy sample with twenty points between 5 and 100 GeV for external evaluation. Particle identification is compared using a boosted decision tree, a multilayer perceptron, a permutation-invariant point-wise network, and a graph neural network. On the common set of 6027 externally evaluated events, their areas under the ROC curve are 0.674, 0.679, 0.655, and 0.649, respectively. The cluster-level networks were trained with a substantially smaller event budget and therefore do not constitute a capacity-matched comparison. For energy reconstruction, a permutation-invariant DeepSets model learns an event-wise correction to the conventional three-threshold calibration. On 100744 contained showers it yields a stochastic term of $(45.5 \pm 1.0)\%$, compared with $(54.4 \pm 1.2)\%$ for the semi-digital calibration, while maintaining a constant term of $(6.4 \pm 0.2)\%$. Finally, explicit routing by the predicted or true particle label does not improve the resolution of an inclusive boosted-tree regressor within the evaluated pion–proton mixture. These results show that cluster-level information improves energy reconstruction in simulation, while the benefit of an explicit pion–proton decision depends on the downstream estimator.

Keywords: highly granular calorimetry; SDHCAL; hadronic showers; particle identification; energy reconstruction; graph neural networks; DeepSets

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1 Introduction

Precise reconstruction of hadronic final states is central to the physics programme of present and future collider experiments. Hadronic calorimetry is intrinsically challenging because invisible energy, fluctuations of the electromagnetic fraction, nuclear breakup, and shower leakage vary substantially from event to event. Particle flow reconstruction mitigates part of this problem by assigning each particle to the subdetector that measures it most precisely, but it requires calorimeters with fine transverse and longitudinal segmentation [1].

The CALICE Semi-Digital Hadron Calorimeter (SDHCAL) combines centimetre-scale segmentation with a two-bit readout. Instead of measuring the full induced charge, each fired pad records the highest of three thresholds that it crosses. This design retains information on local shower density while limiting the readout complexity and data volume [2, 3]. The resulting sparse point cloud is naturally suited both to engineered shower descriptors and to neural architectures that operate directly on unordered sets or graphs.

Recent studies have used graph-based or set-based networks for hadron identification and energy reconstruction in highly granular calorimeters. Graph attention and DeepSets models were compared for a digital calorimeter based on RPWELL chambers, including particle identification and detector-design studies [4]. A dynamic reduction network was also validated on charged-pion beam-test data from a prototype of the CMS High Granularity Calorimeter [5]. The detector technologies, readout schemes, particle samples, and evaluation procedures of those studies differ from the present one; their numerical results are therefore used only as methodological context.

This work addresses two related questions in a simulated SDHCAL. First, how much pion–proton discrimination is obtained from global shower descriptors compared with cluster-level set and graph representations? Second, does an explicit particle label improve energy reconstruction once the regressor already sees detailed shower information? The analysis is restricted to π^- and proton showers. Its main contributions are a common external benchmark for four identification models, a cluster-level energy correction compared with parametric and boosted-tree estimators, and a controlled comparison of particle-blind, classifier-routed, and oracle-routed energy reconstruction.

2 Detector simulation and data samples

2.1 Semi-digital calorimeter model

The detector model follows the CALICE technological prototype. It contains 48 active planes. Each sampling period comprises a 15 mm 304L stainless-steel absorber structure and two 2.5 mm 316L steel cassette skins, for a total of 20 mm of steel. The transverse segmentation is 96×96 pads with a 10.408 mm pitch, corresponding to an instrumented width of 999.168 mm. The active medium is a 1.2 mm glass resistive-plate-chamber gas gap containing 93% tetrafluoroethane, 5% carbon dioxide, and 2% sulphur hexafluoride [6]. Three ordered charge thresholds, $T_1 < T_2 < T_3$,

Table 1: Roles of the two separate Geant4 productions. Counts are per particle species before analysis selections. Individual models can use smaller training budgets, as reported with their results.

Role	Events per species	Energy spectrum	Use
Training/calibration	$\simeq 2 \times 10^5$	continuous, 1–130 GeV	fit and selection
External evaluation	10^5	20 points, 5–100 GeV	final metrics

encode the local activity. The event-level threshold counts N_1, N_2, and N_3 count pads assigned to each exclusive threshold category.

Particle transport is simulated with Geant4 10.04.p03 [7, 8] and the QGSP_BERT reference physics list. Neutrons are retained. The digitisation is run with iLCSoft v02-02-01 and Marlin v01-17-01. It applies a uniform detection efficiency of 0.96, an induced-charge Polya model with mean parameter 5.55942 and width parameter 2.42211, and the exact charge-sharing model with parameter $d = 0.221368$. The three charge thresholds are 0.114, 6.12, and 16.83 pC; the time cut is 1000 ns. These values are recovered from the retained steering code and production logs rather than from metadata embedded in every event file.

Each reconstructed hit carries cell coordinates (I, J, K), spatial coordinates (x, y, z), a threshold category, and a time stamp. A Gaussian time smearing with a standard deviation of 0.1 ns is added only by the machine-learning workflows. This value is an analysis assumption and is not presented as an experimentally validated detector resolution.

2.2 Separate training and evaluation productions

Two separately generated, non-overlapping pion–proton productions are assigned fixed roles, as summarised in Table 1. The first contains approximately 2×10^5 events per species with a continuous energy distribution between 1 and 130 GeV. It is used for model training, internal calibration, early stopping, and fitting the semi-digital energy coefficients. The second contains 10^5 events per species at twenty fixed energies from 5 to 100 GeV in steps of 5 GeV, with 5000 generated events per particle and energy point. It is reserved for external evaluation. Thus, the external set is not obtained by splitting the training production.

For each production, the pion and proton launchers use the same deterministic seed schedule. Consequently, incident positions and directions are matched across species, as are the sampled energies in the continuous production. In addition, a small part of the seed schedule is reused between the continuous and fixed-energy productions. The samples contain distinct simulated events and files, but they are therefore not claimed to be statistically independent random streams.

Event identifiers are rebuilt from the species, input shard, and local event number, because local numbering restarts in each shard. Distinct numerical offsets are assigned to the two productions. These identifiers prevent collisions within the analysis and make prediction tables joinable. They establish event disjointness in the analysis; they do not establish independence of the underlying pseudo-random streams.

2.3 Containment selection

Energy reconstruction is evaluated on showers with no hit in the last active layer. The semi-digital calibration is fitted under this requirement, so applying the same selection to every estimator is necessary for a like-for-like comparison. The selection is strongly energy dependent: its efficiency decreases from 95.8% at 5 GeV to 23.1% at 100 GeV, as shown in Table 2. At 100 GeV, the efficiencies are 24.7% for pions and 21.5% for protons. Consequently, the reported resolution parameters characterise the detector together with this selection, not an inclusive detector response.

Table 2: Fraction of fixed-energy events retained by the longitudinal containment requirement.

E_{true} [GeV]	5	20	50	80	100
Retained fraction [%]	95.8	75.3	48.1	29.2	23.1

3 Event representations and learning methods

3.1 Shower descriptors

The tabular representation consists of 24 event-level descriptors. They include the three threshold fractions; the shower start; transverse radius and density; the number, mean size, and maximum size of intra-layer clusters; longitudinal barycentre and spread; early-layer activity; geometrical eigenvalues and eccentricity; fitted longitudinal-profile variables; reconstructed track segments; and timing summaries. Two exactly or nearly redundant variables used in earlier iterations of the analysis are excluded from the final tabular feature list. The descriptors are computed only from digitised hits.

The boosted decision tree (BDT) is implemented with LightGBM [9]. The multilayer perceptron (MLP) has three hidden layers with 128, 64, and 32 units, ReLU activations, Adam optimisation, and an L_2 penalty of 10^{-4}. For the MLP, the inputs are standardised using scalers fitted only on the training partition; the BDT uses the native descriptor values. Both models use an internal energy- and class-stratified calibration subset of the continuous production for model selection.

3.2 Cluster representation

Neighbouring fired pads are grouped by four-connectivity in (I, J) within each layer K. A shower contains about 753 hits and 191 clusters on average. Each cluster is represented by eight quantities: its three-dimensional centroid, earliest time, fractions of pads in the three threshold categories, and $\log(1 + n_{\mathrm{pad}})$. The spatial coordinates are robust-scaled and time is quantile-transformed using the training sample only.

Two binary classifiers operate on these cluster sets. The first is stored under the historical name *CNN*; it applies two one-dimensional convolutions with kernel size one to every cluster independently, followed by global maximum and mean pooling. It is therefore a permutation-invariant, PointNet-like model [10], not a spatial convolution between neighbouring clusters and not an implementation of the complete original PointNet architecture.

The second classifier is a graph neural network (GNN) [11]. A static k-nearest-neighbour graph with $k = 8$ is built from the scaled cluster features, with a 2 ns time window. A linear encoder is followed by a four-head graph-attention layer [12], two graph-convolution layers, graph normalisation, GELU activations, dropout, and a residual connection. Global maximum and mean pooling feed a binary classifier. The implementation masks candidate neighbours according to time ordering and subsequently symmetrises the selected edges. The graph used by the network is therefore undirected, and no causal interpretation is assigned to it here.

Both cluster-level PID networks are trained with weighted cross-entropy and AdamW at a learning rate of 10^{-3} and weight decay 10^{-4}. The PointNet-like model uses batches of 64, up to 40 epochs, and a patience of eight epochs; the GNN uses batches of 32, up to 30 epochs, and a patience of six epochs. The checkpoint with the largest internal validation AUC is retained.

3.3 Semi-digital and learned energy estimators

The reference estimator is the conventional three-threshold calibration [3],

$$E_{\mathrm{SD}} = \alpha(N_{\mathrm{hit}})N_1 + \beta(N_{\mathrm{hit}})N_2 + \gamma(N_{\mathrm{hit}})N_3, \qquad N_{\mathrm{hit}} = N_1 + N_2 + N_3, \tag{1}$$

where α, β, and γ are quadratic functions of N_{hit}. The nine coefficients are fitted on the continuous production by minimising

$$S = \sum_i \frac{(E_{\mathrm{rec},i} - E_{\mathrm{true},i})^2}{E_{\mathrm{true},i}}. \tag{2}$$

Although this objective is historically denoted χ^2, its denominator is not an event-wise measured variance; it is a weighted least-squares objective. In the particle-blind baseline, pion coefficients are applied to both species.

Two BDT energy regressors use nine shower descriptors: density, cluster count, longitudinal barycentre, mean cluster size, the three threshold counts, total hit count, and the threshold-weighted hit sum. One predicts the energy directly. The other predicts the same multiplicative correction used by the cluster network.

The retained cluster-level energy model is a DeepSets architecture [13], not a GNN. It contains no edges and performs no message passing. Each cluster is encoded into 128 dimensions. Six residual set blocks update each cluster using the concatenation of its state with the event-wide mean state. Global maximum and mean pooling, together with the logarithm of the number of clusters, are passed to the regression head. The network learns

$$y_\theta(G) = \log\left(\frac{E_{\mathrm{true}}}{E_{\mathrm{SD}}}\right), \qquad E_{\mathrm{rec}} = E_{\mathrm{SD}}\, \exp[y_\theta(G)]\, c_{\mathrm{bias}}, \tag{3}$$

where E_{SD} always uses the pion calibration and c_{bias} is fitted on an internal calibration subset. No particle identity is provided to this model. The network is trained with Smooth L1 loss, AdamW, gradient clipping, batch size 128, up to 30 epochs, and early-stopping patience eight. After the containment selection and internal split, 79451 events enter network training.

4 Evaluation protocol

4.1 Particle-identification metrics

Pions are treated as the signal class. The primary metrics are the ROC area and the proton rejection $1/\varepsilon_p$ at fixed pion efficiency ε_π. Unlike accuracy, the latter does not depend on the pion–proton mixture in the evaluation sample. Model predictions are joined by event identifier, and the final four-model comparison uses only events predicted by every model. Statistical uncertainties on AUC differences are obtained from 2000 paired bootstrap replicas, so each replica contains the same resampled events for every classifier.

4.2 Energy metrics

For a fixed true energy, the relative residual is

$$r = \frac{E_{\mathrm{rec}} - E_{\mathrm{true}}}{E_{\mathrm{true}}}. \tag{4}$$

The response is $\mathcal{R} = 1 + \langle r \rangle$. The resolution is the standard deviation of r around its own mean. To reduce sensitivity to non-Gaussian tails, the central 90% interval is retained and its standard deviation is divided by the analytic truncation factor for a Gaussian distribution. This differs from an uncorrected RMS90 and keeps bias and spread as separate quantities.

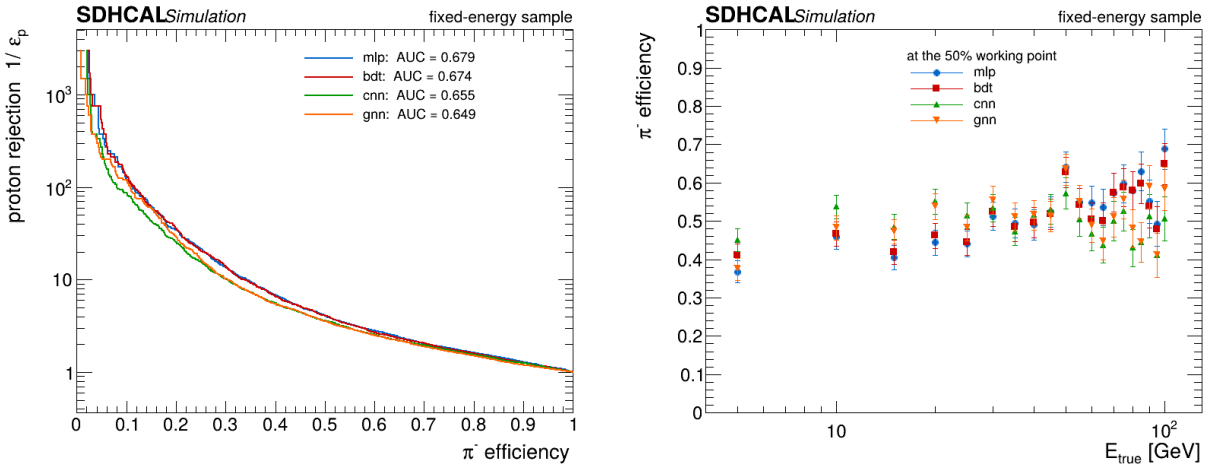
The energy dependence is fitted with

$$\frac{\sigma(E)}{E} = \frac{a}{\sqrt{E/\mathrm{GeV}}} \oplus b. \tag{5}$$

No electronic noise hits are injected by the digitisation used here, so a c/E term is not fitted. Statistical errors in Equation (5) are obtained from the per-energy resolution uncertainties. Systematic uncertainties from detector modelling are not included.

Table 3: Binary pion–proton identification on the common external event set. Rejection is quoted at 50% pion efficiency.

Model	Input	Training events	AUC	$1/\varepsilon_p$
MLP	shower descriptors	176000	0.679	4.07
BDT	shower descriptors	176000	0.674	4.10
PointNet-like	clusters	8400	0.655	3.62
GNN	cluster graph	8400	0.649	3.54



(a) Proton rejection versus pion efficiency.

(b) Pion efficiency versus true energy at the global 50% working point.

Figure 1: External pion–proton identification performance. The internal plot label "cnn" denotes the point-wise permutation-invariant model described in Section 3.2; it is PointNet-like and does not convolve neighbouring clusters.

5 Particle-identification performance

The common external subset contains 6027 events. The tabular models were trained on 176000 events, whereas each cluster-level network was trained on 8400 events. The performance comparison in Table 3 and Figure 1 is therefore an evaluation on common events but not a training-budget-matched architecture study.

The MLP and BDT have nearly identical rejection at the 50% working point. Their AUC difference is 0.0053 ± 0.0022 from the paired bootstrap. The BDT exceeds the PointNet-like and GNN models by 0.0188 ± 0.0047 and 0.0252 ± 0.0051, respectively. The difference between the two cluster-level models is 0.0064 ± 0.0048 and is not resolved by this sample. The pion efficiency increases only gradually with energy and the per-energy statistical uncertainties are too large to establish a different model ordering in individual bins.

Pion–proton separation is expected to be difficult because the electromagnetic fraction and topology fluctuate strongly in pion showers, while baryon-number conservation modifies the secondary-particle composition of proton showers [14]. The present results demonstrate substantial overlap for the evaluated models and data budget. They do not establish a model-independent physical upper bound: the cluster-level models use fewer training events, only one seed is available for each, and no capacity-matched ablation was performed.

Table 4: Energy performance on the common contained external sample. "Correction" denotes the target in Equation (3). The pooled σ/E depends on the flat generated spectrum and is not an intrinsic detector resolution.

Estimator	Training events	a [%]	b [%]	pooled σ/E	max $\lvert\mathcal{R}-1\rvert$
DeepSets correction	79451	45.5 ± 1.0	6.4 ± 0.2	0.1115	0.028
BDT correction	176000	53.7 ± 1.0	5.6 ± 0.3	0.1197	0.052
BDT direct energy	176000	53.6 ± 1.1	5.6 ± 0.3	0.1199	0.054
Semi-digital calibration	—	54.4 ± 1.2	7.0 ± 0.3	0.1295	0.031

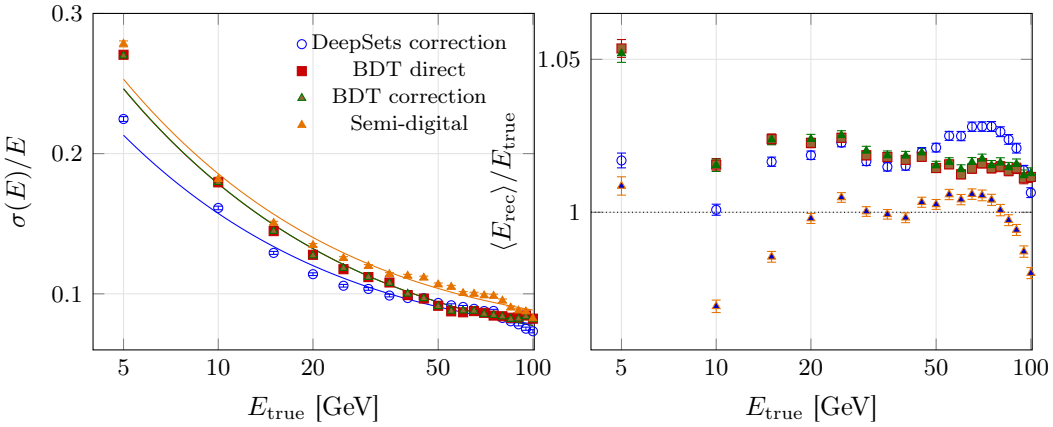


Figure 2: Energy resolution (left) and mean response (right) on the fixed-energy sample. Error bars are statistical. The curves in the resolution panel show fits to Equation (5). The source artifact historically named `gnn_ratio` is labelled DeepSets here because the retained run uses no graph edges.

6 Energy-reconstruction performance

All energy estimators are compared on the same 100744 contained events from the fixed-energy production. Table 4 reports the fitted stochastic and constant terms, the resolution obtained after pooling this particular flat grid of generated energies, and the largest response deviation. The pooled number is sample-spectrum dependent and is included only as a compact cross-check; a and b are the primary figures of merit.

The BDT variants and the semi-digital calibration have compatible stochastic terms near 54%. The improvement of the BDT over the calibration is concentrated in the constant term, which decreases from $(7.0 \pm 0.3)\%$ to $(5.6 \pm 0.3)\%$. Changing the BDT target from direct energy to the correction of Equation (3) has no significant effect.

The DeepSets correction instead decreases the stochastic term to $(45.5 \pm 1.0)\%$. It gives the best resolution over the complete evaluated range and keeps the maximum response deviation below 2.9%. Because DeepSets contains no graph edges, this gain cannot be attributed to explicit cluster connectivity. It is evidence that the cluster-level representation and learned aggregation contain useful event-wise information beyond the threshold counts and selected global descriptors.

A control model trained directly on $\log E$ reached a pooled resolution of 0.203 and a stochastic term near 78%, while its response changed from approximately 1.21 at 5 GeV to 0.81 at 100 GeV. This regression-to-the-mean behaviour motivates the ratio target in Equation (3). Conversely, applying the ratio target to the BDT does not improve its result, which indicates that the DeepSets gain is not explained by the target transformation alone.

Table 5: Global mean bias and resolution for the coupled PID–energy study. All scenarios use the containment selection. Predicted-label rows contain 100678 events; the other rows contain 100744.

Label source	Energy estimator	Mean bias [%]	σ/E
none (pion reference)	semi-digital	-1.36	0.1342
predicted BDT label	semi-digital	$+0.60$	0.1287
true label	semi-digital	-0.27	0.1295
none (inclusive)	BDT	$+2.00$	0.1199
predicted BDT label	BDT	$+2.02$	0.1201
true label	BDT	$+1.88$	0.1199

7 Does explicit particle identification improve energy reconstruction?

The downstream effect of particle identification is isolated by crossing three label sources with two estimator families. The label is either absent, supplied by the BDT classifier, or set to the true species as an oracle. Without a label, the parametric method uses pion coefficients for every event, whereas the learned method uses one inclusive BDT trained on the pion–proton mixture. With a predicted or true label, species-specific parametric calibrations or BDT regressors are selected.

Species routing improves the parametric result relative to applying pion coefficients to all events. This comparison mainly corrects a mismatched proton calibration and should not be interpreted as an absolute gain from PID. Once the baseline is an inclusive BDT trained on the mixture, no improvement is observed: the resolutions without a label, with the predicted label, and with the true label are 0.1199, 0.1201, and 0.1199. Their statistical uncertainties are approximately 3×10^{-4}, and the inclusive BDT has a seed-to-seed spread of 0.0008 over five seeds.

In this setting, the global shower descriptors already contain the information that the explicit species label would provide. Routing replaces this continuous information with a hard binary decision and therefore adds no measurable resolution. This result is conditional on the balanced pion–proton mixture, the simulated detector, and the trained energy range. It need not hold for a strongly shifted particle composition, an unseen species, or detector data with different feature resolutions.

8 Discussion and limitations

The study supports two distinct conclusions. First, under the available training budgets, global shower descriptors are at least as effective as the tested cluster graph for pion–proton identification. The unequal budgets prevent attributing the difference solely to representation or architecture. Second, cluster-level information does improve energy reconstruction when used by the DeepSets correction. This finding does not demonstrate a benefit from graph connectivity; a direct, budget-matched comparison between DeepSets, a static graph, and a dynamic graph remains necessary.

Several limitations define the scope of the numerical results:

- all samples are simulated, with no beam-test validation or detector-model systematic uncertainty;
- the simulation configuration was recovered from copied build files, launch scripts, and logs rather than from immutable metadata embedded in the event files; the production scripts and local extraction changes must be archived with content hashes before submission;
- pion and proton jobs share deterministic seed schedules, and the digitisation jobs reuse a fixed random seed; the effect of the resulting correlations has not been quantified;

- parallel digitisation jobs contended for a shared auxiliary AIDA output. The logs report successful LCIO completion for all 40 fixed-energy shards per species, but the diagnostic-file write failures should be eliminated in any reproduced production;
- the 0.1 ns timing smearing is assumed, and no timing ablation is included in the retained benchmark;
- the containment selection changes from 95.8% to 23.1% across the energy range and preferentially retains compact high-energy showers;
- the DeepSets result corresponds to one training seed, whereas a five-seed spread is available only for the inclusive BDT;
- the PID comparison uses a model-dependent common intersection of 6027 events and markedly different training budgets;
- the quoted uncertainties are statistical and do not cover hadronic-model, calibration, efficiency, threshold, noise, or alignment variations.

These limitations are particularly important when comparing with published DHCAL or HGCAL results. Differences in absorber depth, active medium, analogue versus digital readout, leakage treatment, particle composition, and core-width definition preclude a direct numerical ranking [4, 5].

9 Reproducibility

The analysis keeps the continuous production for training and internal calibration and the fixed-energy production for external evaluation. Geant4 10.04.p03 and QGSP_BERT are confirmed by the retained fixed-energy production logs and the copied executable build tree; all four particle-production launchers select the same physics list. Analysis random seeds, feature order, event identifiers, containment flags, model metadata, and event-level predictions are recorded by the software workflow. The simulation seed schedule is retained in the launch scripts but is not embedded in the event files.

The article tables are frozen from the common benchmark outputs, and the energy curves in Figure 2 are stored as plain-text data beside this source. The software repository URL, an immutable archive of the exact local simulation and extraction sources, and an archival DOI will be added before submission. The simulated event samples are not included in the source repository; their access and preservation policy must be agreed by the authors and the relevant collaboration.

10 Conclusion

Binary pion–proton identification and hadron energy reconstruction have been studied with separate training and fixed-energy evaluation productions in a simulated semi-digital calorimeter. On a common external subset, an MLP and a LightGBM classifier using shower descriptors reach AUC values of 0.679 and 0.674. A PointNet-like cluster model and a static GNN reach 0.655 and 0.649 with twenty times fewer training events; the comparison therefore motivates, but does not replace, a budget-matched study.

For energy reconstruction, a particle-blind DeepSets correction to the three-threshold calibration reduces the fitted stochastic term from $(54.4 \pm 1.2)\%$ to $(45.5 \pm 1.0)\%$ on contained showers. The corresponding constant term is $(6.4 \pm 0.2)\%$. An inclusive boosted-tree regressor is not improved by adding either the predicted or true pion–proton label, showing that explicit PID routing is redundant for that estimator in the evaluated mixture. Validation on beam-test data, simulation systematics, repeated neural-network training, and controlled comparisons of set and graph architectures are required before drawing an experimental conclusion.

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